

DIGITAL LOGIC

Case Study - UART

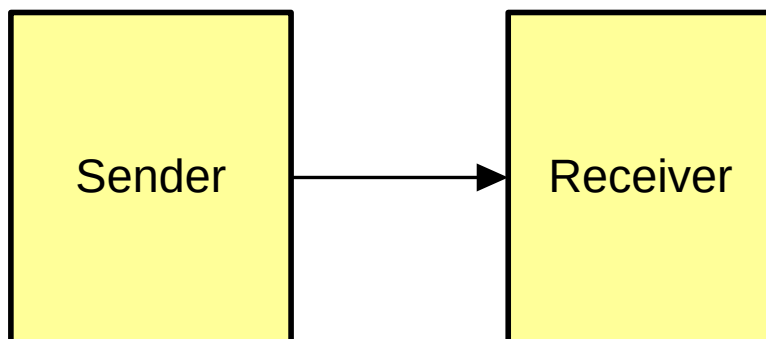
2025 Fall

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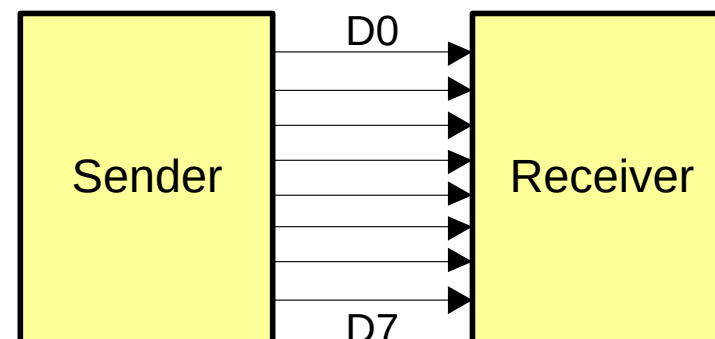
Serial vs. Parallel Communication

Characteristics	Transmission Rate	Anti-Interference Ability	Communication Distance	I/O Resource Usage	Cost
Serial Transfer	Low	High	High	Low	Low
Parallel Transfer	High	Low	Low	High	High

Serial Transfer

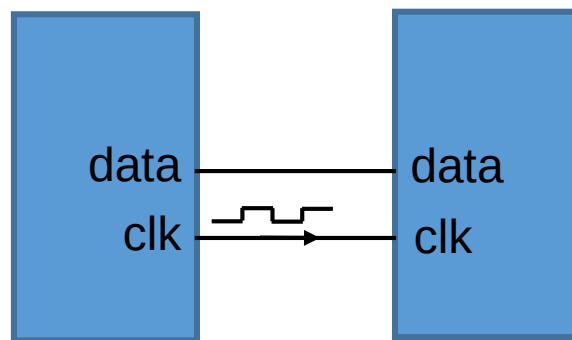


Parallel Transfer

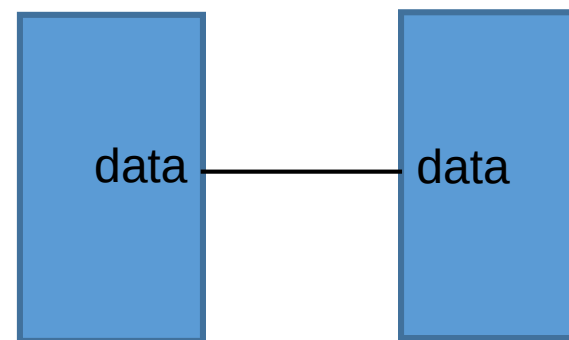


Synchronous vs. Asynchronous Communication

- Synchronous Communication
 - Shares the same clock signal, which is sent along with data;
 - The device that generates the clock is called the **master** and other devices are **slaves**
- Asynchronous Communication
 - no clock transmitted, fewer wires
 - it relies on synchronous signals like start and stop bits within the data signal



Synchronous



Asynchronous

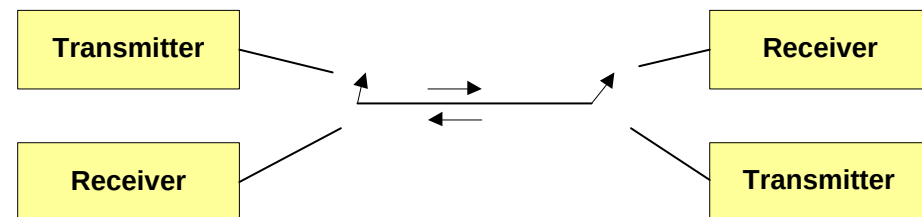
Direction of Communication

- Simplex Communication:
 - Data can only be transmitted in one direction
- Half-Duplex Communication
 - Data can be transmitted in both directions but need time division
- Full-Duplex Communication
 - Data can be simultaneously transmitted in both directions

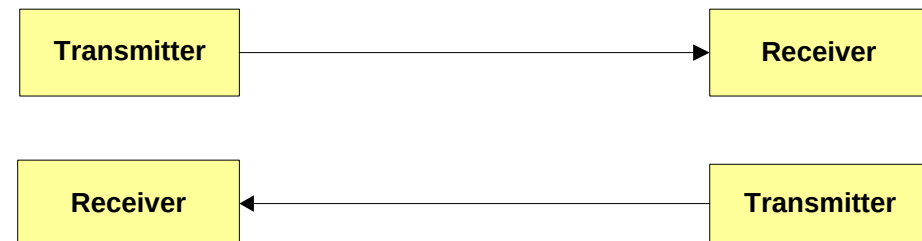
Simplex



Half Duplex

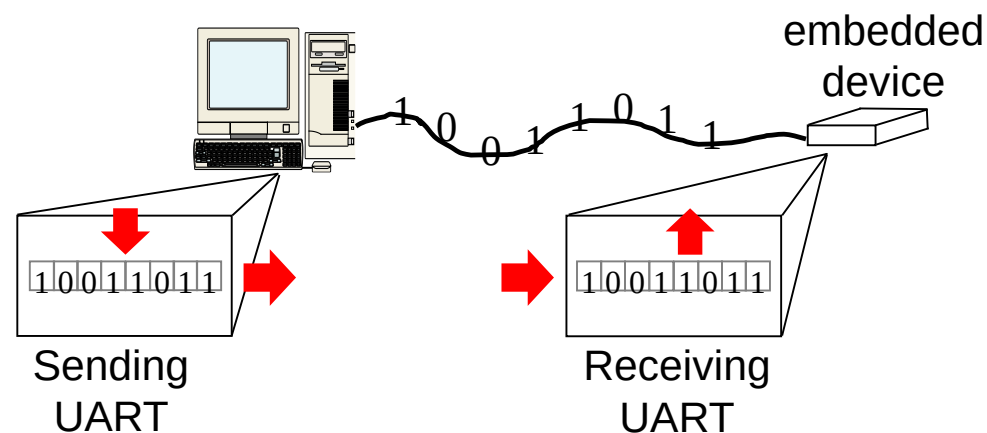


Full Duplex

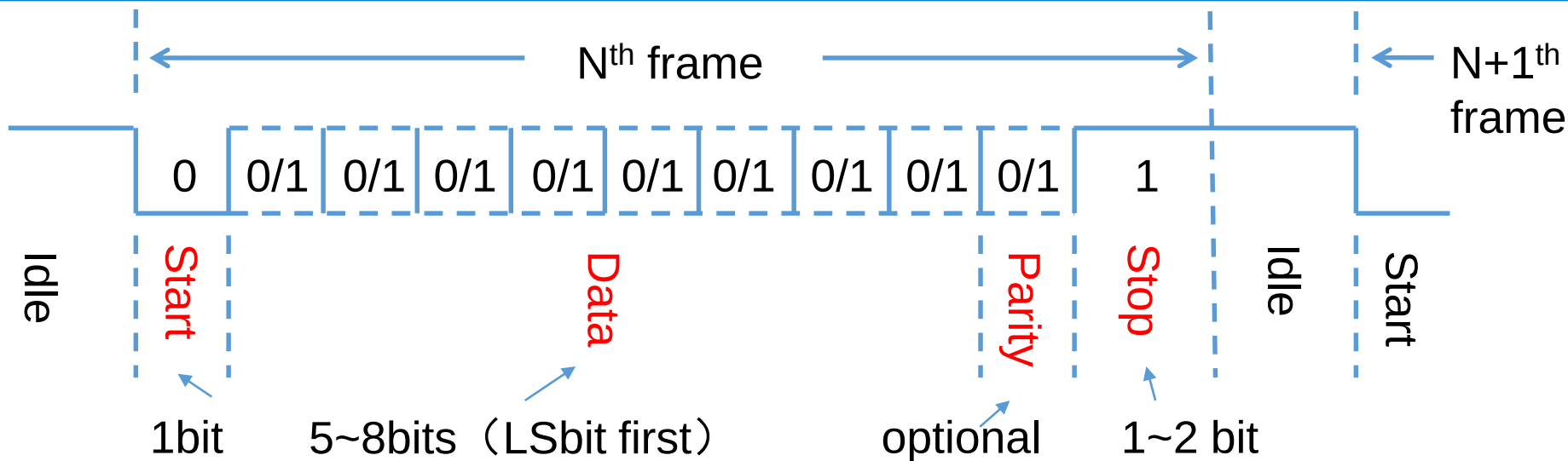


UART

- UART (Universal Asynchronous Receiver-Transmitter)
 - a universal **serial asynchronous** communication bus with two data lines, enables **full-duplex** transmission and reception, and is commonly used in embedded systems for communication between a host and peripheral devices.
- How to synchronize the transmissions of the two ends which run on independent clocks?
 - Use absolute (real) time
 - We use: Transmit short data (e.g. one byte) at a time, assuming the two clocks run at same rate during that period of time



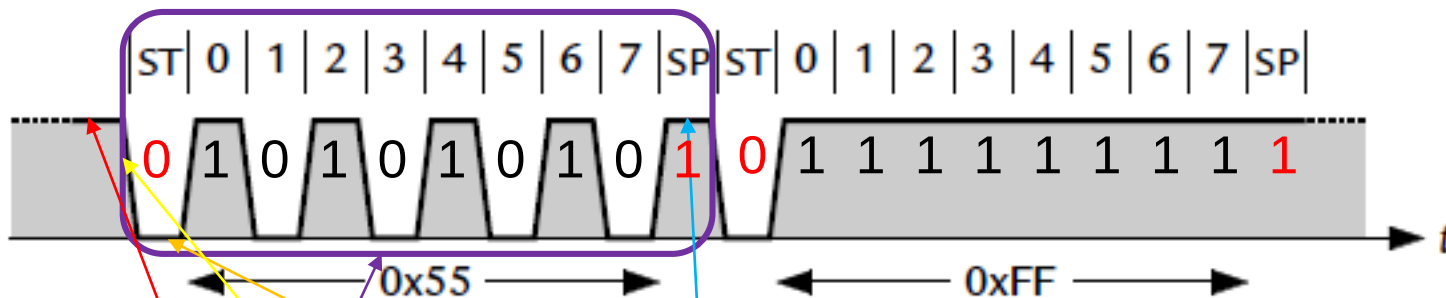
Frame Format



- Data are sent in short frames, each of which typically contains a single byte
- Data frame
 - One logic-low start bit
 - Data (**LSB first**, and size of 5~8 bits)
 - One optional parity bit to make total number of ones in data even or odd
 - One or two logic-high stop bits

Frame Example

- Example: 8-N-1 format: 8-bit data, no parity bit, 1 stop bit



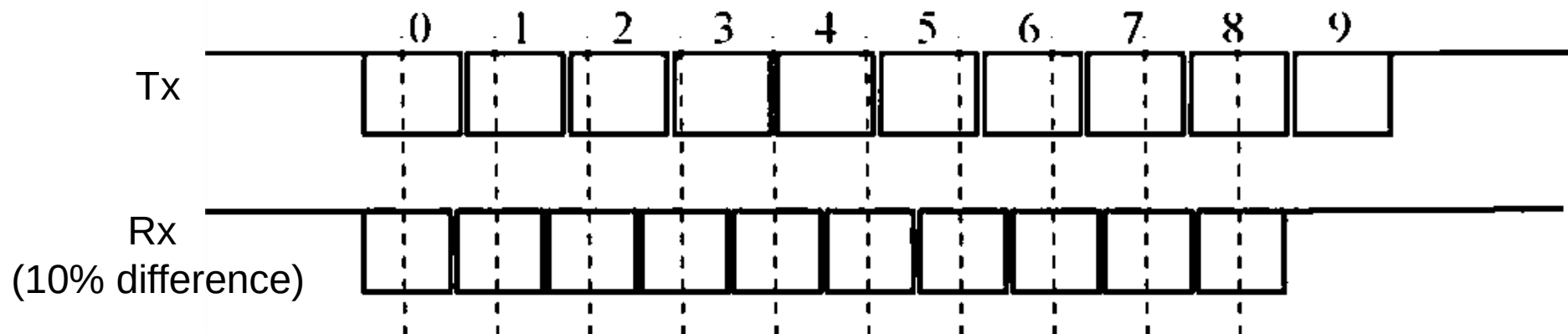
- For each frame transmission:
- The line is high when no data is sent
- The transmitter sends a 0 bit as a start bit (ST)
- The receiver uses the falling edge to be synched with the sender
- Then data 0x55 is sent (LSBit first)
- The line becomes high for 1 stop bit (SP) to make sure that the next start bit makes a falling edge
- sequence sent is 010101011

Transmission Speed

- Receiver must know the transmission rate – Baud Rate(波特率)
 - # of signal changes per second,
 - e.g., 9600 、 19200 、 38400 、 57600 and **115200** baud (bps)
 - In UART, each signal change represents one bit, so baud rate (Baud) and bits per second (**bps**) are equal
 - 8-N-1 format, Since each 8 bits of data are accompanied by a start and a stop bit, maximum data rate is only 8/10 of baud rate
- The baud rate of the transmitter and receiver must be consistent to ensure the transmission.
 - e.g.
 - 50 MHz frequency: $50\text{M} / 115200 = 434$ clock cycles to transmit/receive 1 bit.
 - 100 MHz frequency: $100\text{M} / 115200 = 868$ clock cycles to transmit/receive 1 bit.

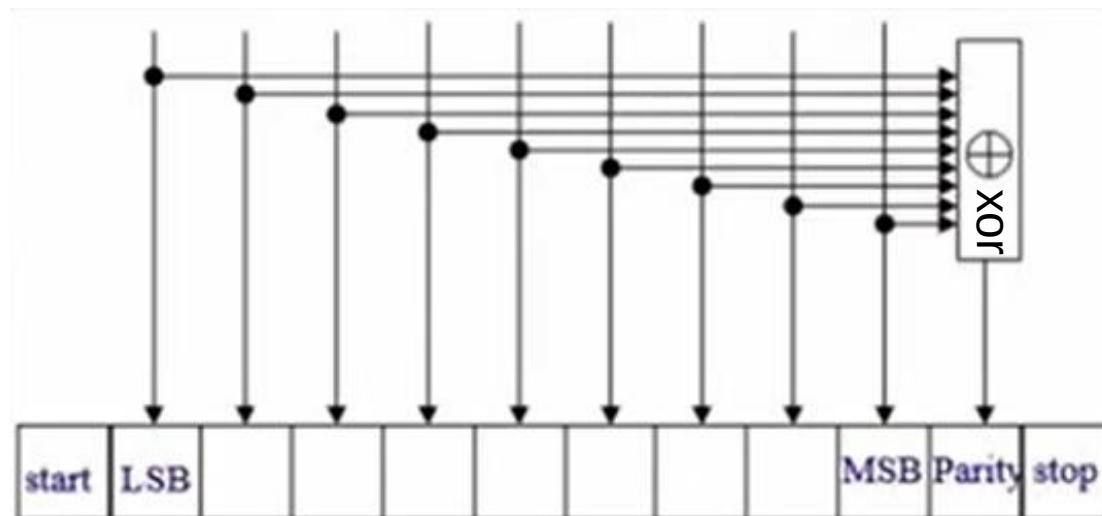
Synchronization of two ends

- Transmitter and Receiver have their own clocks, and they might be different in phase.
 - The data is sampled at the middle of each clock cycle
 - E.g. 10% difference might lead to transmission error for 8-N-1 format
- **Divide transmission into small data amount:** e.g. data frame of 8-N-1 format

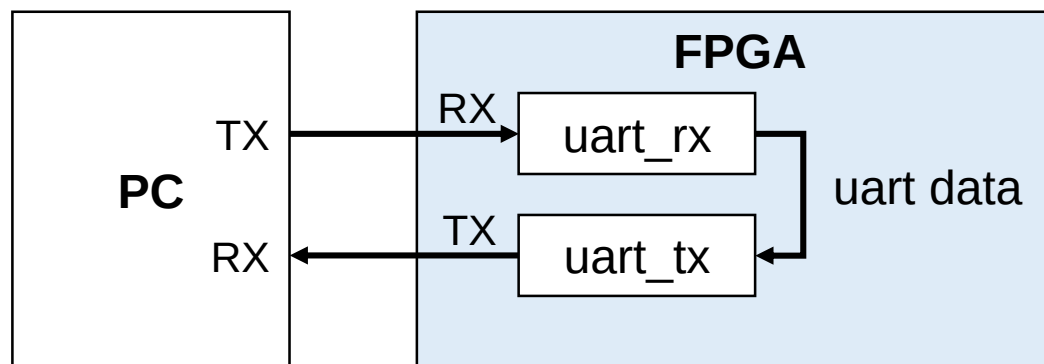


Error Detection

- Problem: How to check the data integrity?
- Solution: Parity bit is added to the tail of frame.
 - Even Parity: total number of “1” bits in data and parity is even
 - Odd Parity: total number of “1” bits in data and parity is odd
 - Example: Data = 10101011 (five “1” bits)
 - The parity bit should be 0 for odd parity and 1 for even parity
 - This can detect single-bit data corruption



Architecture for loopback uart

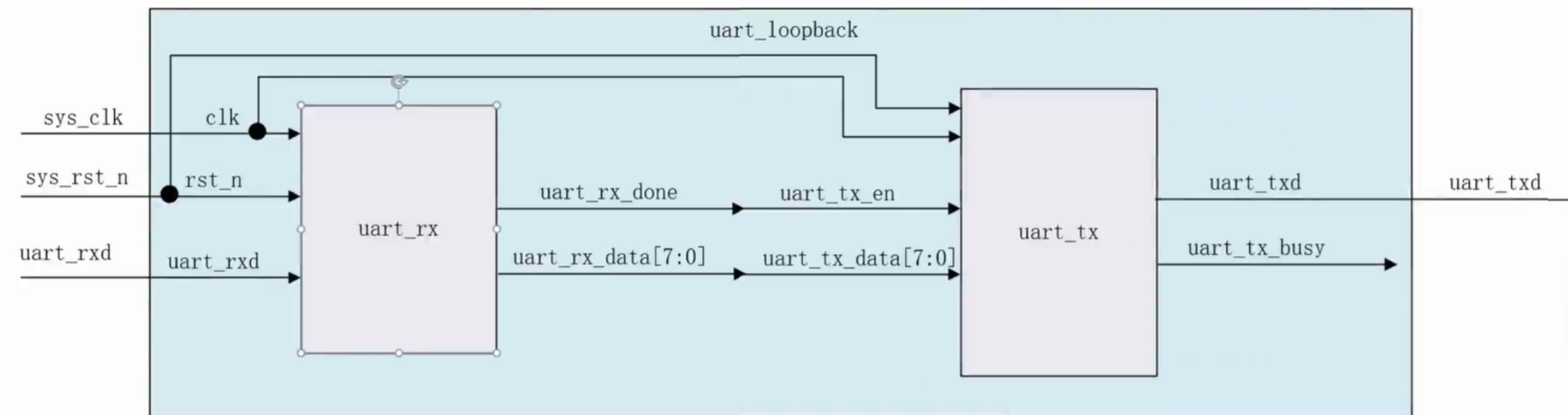


example by 正点原子
available on BB

 uart_loopback

 uart_rx

 uart_tx

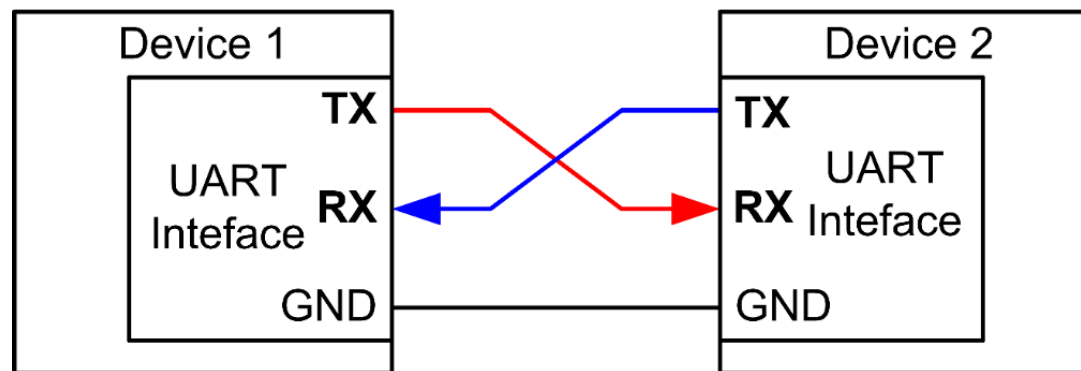


正点原子 Online video:

https://www.bilibili.com/video/BV1E94y1p7jk?spm_id_from=333.788.videopod.episodes&vd_source=027f3f85977403d2caf9e525371700c8&p=78

UART Connection

- Universal
 - UART is programmable.
- Asynchronous
 - Sender provides no clock signal to receivers



UART physical implementation

- Problem of directly using UART
 - UART is a communication mechanism, it only defines the timing sequence, but does not specify the electrical characteristics of the interface.
 - When using UART communication, processors typically use TTL levels. However, there are differences in the voltage levels used by different processors, so UART connections between different processors usually cannot be directly connected.
 - UART does not specify a standard for connectors when different devices are connected
- In practice, we use RS232, which defines the electrical and mechanical characteristics of the interface

RS-232

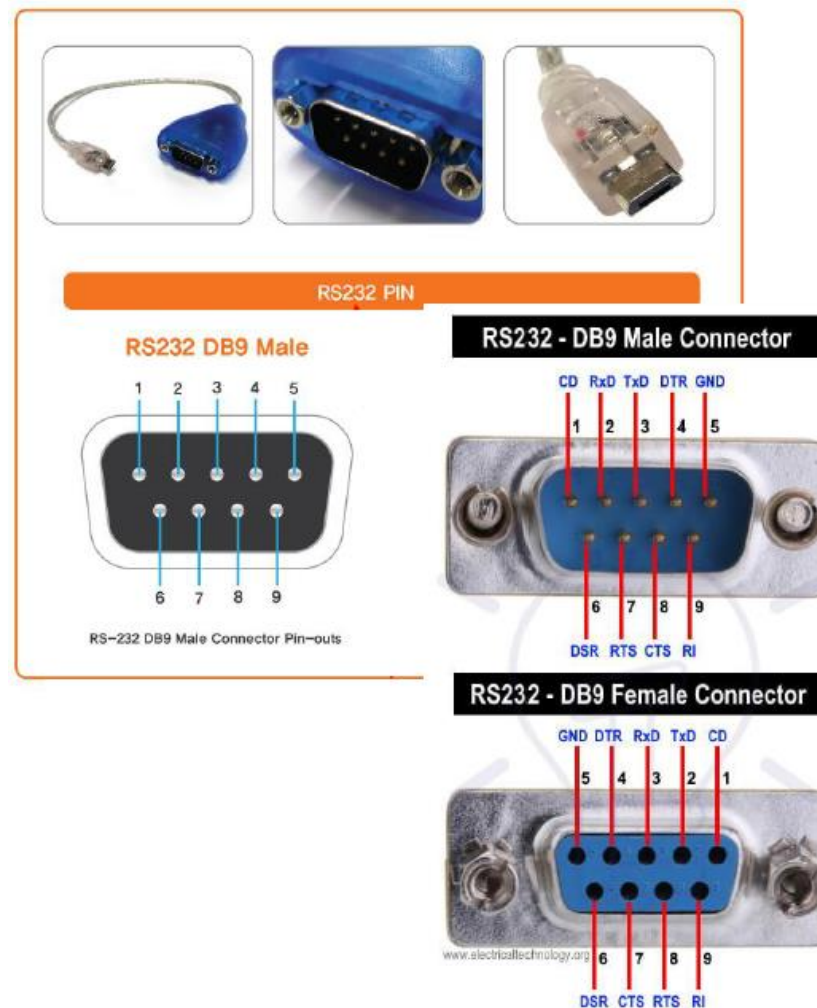
- RS232 defines the electrical and mechanical characteristics of the interface for serial communication.
 - while UART has everything to do with logic and programming, but RS232 refers to the electronics and hardware needed for serial communications

Standard	Voltage signal	Max distance	Max speed	Number of devices supported per port
RS-232	logic 1: -15V to -3V, logic 0: +3 to +15 V)	50 feet	20Kbit/s	1 master, 1 receiver
RS-485	Differential (-7V to +12V)	4000 feet	10Mbit/s	32 masters, 32 receivers

RS-232 DB9 Connector

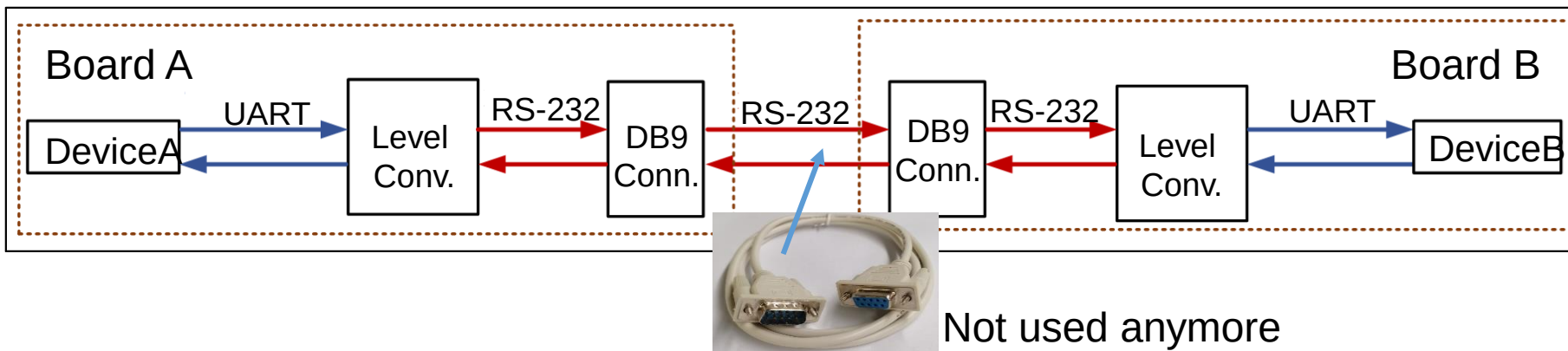
- 9 pins

Pin	Description
1	Data carrier detect (DCD)
2	Received data (RxD)
3	Transmitted data (TxD)
4	Data terminal ready (DTR)
5	Signal ground (GND)
6	Data set ready (DSR)
7	Request to send (RTS)
8	Clear to send (CTS)
9	Ring indicator (RI)



UART Between Devices

- A level converter chips converts UART default TTL voltage level to RS-232 voltage level



- USB to UART converter adapts UART port to a standard USB interface

